Fusionista: Fusion of 3-D Information of Video in Retrieval System

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Abstract. This paper presents Fusionista, an interactive video retrieval system developed for the Visual Browser Showdown 2025. Utilizing CLIP-based feature extraction, Fusionista captures comprehensive visual representations to boost search performance. It also integrates multiple modalities, including color information, textual information, image similarity, and speech information. Additionally, we introduce fusion search, which divides and combines various search functions for efficiency, and temporal search, which identifies the best frame sequences. The user interface is also optimized for fast, intuitive navigation, making it user-friendly.

Keywords: Video Browsing · Context-based Retrieval · Multimodel Representation · Text-Image Interactive Retrieval

1 Introduction

As videos play a central role in digital interactions, the need for efficient video retrieval systems is increasing. The annual VBS aims to benchmark interactive video retrieval systems through live, metrics-based evaluations. Participants are tasked with three categories: Known-Item Search (KIS), Ad-hoc Video Search (AVS), and Question Answering (QA). KIS is divided into text-based (t-KIS: using textual queries) and visual-based (v-KIS: using visual features) to find target images or videos. Besides, AVS allows users to pose broad queries without exact content. In 2024, QA was introduced, requiring the system to answer specific questions based on video content, adding complexity to the retrieval

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process. The VBS2025 challenge data combines more than 4800 h of V3C video content, extended marine video, and the LapGynLHE surgical dataset.

In order to address all three VBS tasks, our system processes inputs (user queries and a repository of information) and outputs the top k most relevant results. We divided the process into three stages: data preparation and extraction, system processing (Back-end), and display processing (Front-end). First, video data is segmented into keyframes for analysis, with information extracted from image, audio, and temporal dimensions. This data is then processed and queried by Elastic Search (ES) for non-vector data and FAISS [2] for vector data. We implemented the back-end services by FastAPI to process search functions and retrieve data from the database. Then, the results are displayed with reduced resolution for faster loading on a user interface built with ReactJS Tailwind. Our system enhances the user experience, particularly for novices, by offering various input methods (text, audio, and visual) and supporting fast, detailed, and consecutive search options. Additionally, our novel search function allows combining multiple features with long queries (e.g., object-color-location or text-image) to refine searches effectively.

2 Data Preparation and Extraction

We process the videos provided by the organizers by separating them into audio and video components. For audios, we use a speech-to-text model to generate transcripts, which are stored with timestamps for future use. We convert video processing into image processing by extracting keyframes to improve the accuracy of the video component. The keyframe extraction method is taken from the study of Tan et al. [16], where the entire video is embedded into vectors using the CLIP (B-32) [12] model for its speed and performance. In contrast, TransnetV2 [15] is used for scene boundary detection. We cluster vectors within each scene to select representative keyframes. Then, we extract data (including text-image embeddings and object detection) to support developing specialized search functions for several aims.

3 User Interface

The Fusionista is a user-friendly web application (Fig. 1) developed to create and execute queries across multiple retrieval modalities. It presents the top query results through an intuitive interface, enabling users to browse and explore the content easily. Users can view the corresponding videos, interact with the search results, and submit the result by simply clicking or pressing Enter.

The UI in Fig. 1 consists of six panels, each corresponding to a different search modality: semantic, temporal, OCR, ASR, object-color detection (inspired by VISIONE interface [1]), and QA. These options offer diverse ways for users to refine their queries and retrieve relevant information.

Each image result includes multiple buttons or functionalities, such as image similarity search, viewing the associated video, previewing 40 images around the

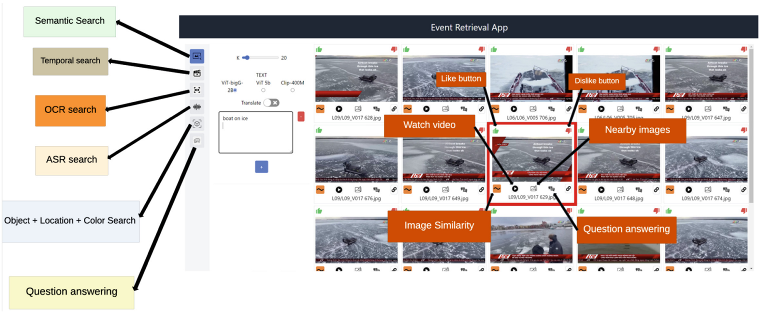

Fig. 1. Main User interface.

selected one, and a VQA button for question-answering tasks. Above the images are "*like*" and "*dislike*" buttons, which allow users to provide feedback on the images. This feedback directly contributes to refining the semantic search results.

4 System Key Functions

In this section, we will introduce the system's main functions described in Fig. 2, including the basic functions of the query systems inherited from VBS [1,10,13] in previous years. We also explain our fusion search and temporal search functions with new similarity functions to support the retrieval results better. In addition, we also use Elastic Search [6] for functions related to object search to increase speed.

4.1 Textual Search

Semantic Search with Translation and Query Refinition: We use multiple versions of the CLIP [12] models to obtain semantic embeddings for the extracted images. To enrich our system's capabilities in supporting multimodal search, we used ViT-H-14, ViT-SO400M-14-SigLIP-384, ViT-bigG-14, and PubMedCLIP [3] for medical data in LapGynLHE to generate image embeddings. These embeddings are then indexed using FAISS [2] for efficient similarity searches. To support users from diverse linguistic backgrounds, we integrate GPT-4o [9], enabling automatic translation of queries from various languages into English, the system's primary processing language. Leveraging the few-shot learning capabilities of GPT models, we further prompt the model to refine and clarify the query. This process ensures that the query better captures the user's intent and is suitable for our image-text models.

OCR/ASR Search: We developed a search feature based on textual information present in keyframes due to its high effectiveness for searching text within images. We used PaddleOCR [7] to extract textual content from keyframes. We

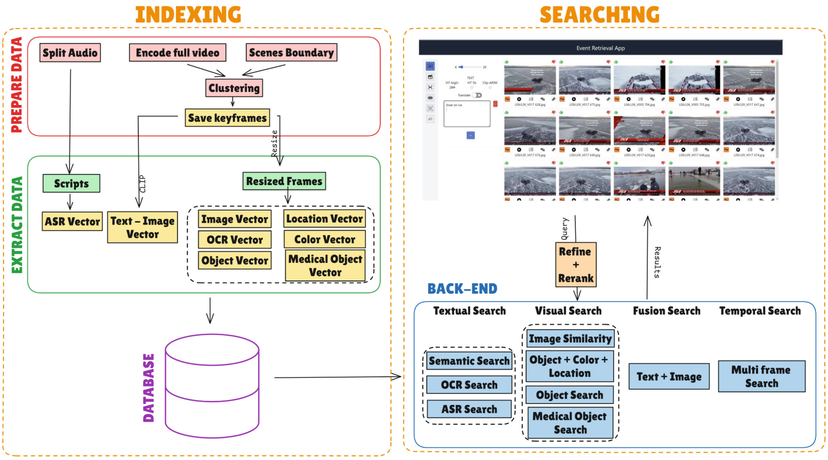

Fig. 2. Overview of system with three main steps.

also implemented two search methods: the first one is linear search, which compares the query with each text sample extracted using the Levenshtein distance [4]. This approach can handle minor errors, such as typographical or spelling mistakes, but is relatively slow; another is ES [6], which leverages a distributed architecture and robust indexing system to store and search data efficiently. However, in some cases, this approach involves a trade-off regarding accuracy.

With ASR, we first extract transcripts with time stamps from the audio using the pretrained Whisper-V3 [11] model, with some modifications for more precise timestamps. We then record all keyframe identities associated with the text, a crucial step in our process and use the same search function with OCR.

4.2 Visual Search

Image Similarity: We observed that many retrieved images and videos exhibited high similarity to one another. To address this redundancy, we implemented an image similarity feature. In this approach, the selected image is encoded using one of the three aforementioned OpenCLIP [12] models. The resulting embedding is then used to perform a similarity search via the FAISS index [2], ensuring that the system retrieves more diverse and relevant results.

Object Number Search: We utilized YOLOv8 [5] for object detection and MedYOLO [14] for recognizing medical objects in images due to its high speed. The number of objects in each image is encoded into a k-dimensional vector (reflecting the number of classes) $V \in N^k$. Each element v_i of vector V represents the number of objects in class i within image I. We implemented multiple search methods. The first method is Fast search, in which all k-dimensional vectors are structured into a list of k sublists, each corresponding to a YOLOv8 class. Each sublist contains a set of information S_j^i, with j representing the number of objects in class i in an image. For query "5-person 2-car" we retrieve sets of keyframes for each object (e.g., "5-person," "2-car," etc.), then intersect these sets to find keyframes that match precisely the number of objects. This method is fast and precise but does not handle slight discrepancies.

We also use ES [6], in which we transform object counts into textual format (e.g., "3-person 2-dog" becomes "person___III + dog___II" where "person" is the name of the class, "___" serve as padding and "III" is the number of objects in the class) then search. This approach is quicker and allows more flexibility in object counts, though it sacrifices some accuracy.

Object - Color - Location Combination Search: In this method, we applied a 7×7 grid (columns $\in$ [a, b, c, d, e, f, g] and rows $\in$ [1–7]) over each keyframe to encode object positions. Each object is represented as a text string combining its grid location and class name (e.g., "a1person" with "a1" being location and "person" as the class of object), with all object strings concatenated to form a full image description. For color encoding, we used a 16-color palette and proposed a hierarchical K-means clustering to map the pixel colors to this space, extracting dominant colors and encoding them similarly to objects. The object and color text strings are then combined to represent each keyframe. We refer to the idea from VISIONE System [1].

Users can create a 7×7 canvas sketch by dragging objects and color labels to simulate the desired image, encoded similarly to the keyframes.

4.3 Fusion Search

Although the pre-trained CLIP model [12] is effective for image retrieval, crafting the right query remains challenging. The query must be detailed yet concise, as CLIP struggles with lengthy text. To mitigate this, we use a similarity method incorporating multiple smaller queries, allowing the model to process each query individually. Specifically, we calculate the similarity between an image and various queries using the exponential mean of their similarities:

$$Score(Q, I) = mean_{q \in Q}(exp(sim(q, I))) \tag{1}$$

where Q is a list of queries and I is an image. $sim(q, I)$ represents the similarity between query q and image I.

While a simpler mean of similarities could speed up retrieval using vector indexing (e.g., FAISS), the exponential mean acts like a softmax function, prioritizing images that closely match one query while favoring images that match all queries. It also aligns better with CLIP's training loss. By supporting multiple queries, users can mix different types of queries, including images and negative queries (like and dislike from human feedback), returning results that are least similar to some queries while most similar to others.

4.4 Temporal Search

Semantic search and, subsequently, Fusion search are made for the task of image retrieval and, therefore, do not leverage the temporal properties of each keyframe (e.g., it considers all images to be independent of each other). It is essential for the task of event retrieval in videos. To address this problem, we allow the user to input multiple queries with temporal property (each query describes a frame that happens consecutively within a video). Then, we would return the video that best matches those queries (and the frames that match those queries within that video). The formula can be written as:

$$Score(Q, V) = max_{(i_0,i_1,\ldots,i_k)\in\mathcal{I}}(sim(Q_0, V_{i_0}) + \ldots + sim(Q_k, V_{i_k})), \quad (2)$$

where Q and V are sequences of queries and frames respectively, $\mathcal{I}$ is the set of all possible frame indices $(i_1, i_2, \ldots, i_k)$ such that $1 \leq i_1 < i_2 < \ldots < i_k \leq n$. and $sim(Q_0, V_{i_0})$ denote the similarity between the first query and the frame at the index i_0 within the video V.

The result is found by checking through all combinations of frames corresponding to the queries within a video and returning the one with the highest mean of similarities. Through dynamic programming, it can be done with a runtime complexity of $\mathcal{O}(n \cdot f)$ (with n and f being the number of queries and the number of frames, respectively).

4.5 Question Answering

Inspired by OpenAI's ChatGPT [9] interface, we developed a chat-based system where users can interact with the system via text. In the VisualQA task, the user's initial query triggers a semantic search, after which they can select an event. Using BEIT-3 [17], and BLIP2 [8], our system generates answers from both text and images. We applied a medical QA approach for surgical videos in the LapGynLHE dataset using PubMedCLIP [3], a fine-tuned version of CLIP trained on PubMed articles. We extended this to VideoQA, which is more complex due to the spatio-temporal nature of videos [18]. The task involves retrieving video segments for spatio-temporal analysis using a VideoQA model.

5 Conclusion

In this paper, we introduce Fusionista, an information retrieval system capable of handling various input types. The system can search for information using images, audio (converted to text), and textual queries. A key feature of our approach is the fusion of multiple short queries to describe a long query, enabling us to handle the limit of CLIP models and re-rank results to better align with user information needs. The system can also use the temporal information to find not only one frame but also a sequence of frame. Additionally, we emphasize the user interface, ensuring it is intuitive and easy to use for novice users.

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